Enterprise Node Orchestration for ZK Validium: Pipelined State Machine Design and Performance Analysis

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***Abstract*—Zero-knowledge validium nodes for enterprise blockchain require orchestrating a multi-phase pipeline—transaction ingestion, Sparse Merkle Tree state updates, witness generation, ZK proof computation, data availability committee attestation, and L1 settlement—while maintaining continuous availability and zero data leakage. We present the design and empirical evaluation of a pipelined, event-driven state machine orchestrator for the Basis Network enterprise ZK validium node on Avalanche. The orchestrator implements six states with 17 valid transitions, enabling concurrent transaction ingestion during proof generation and L1 submission through three decoupled processing loops. Benchmarks across five configurations with 30–50 replications each demonstrate that orchestration overhead (batch formation plus witness generation) consumes 593 ms per 8-transaction batch—just 0.66% of the 90-second end-to-end budget. ZK proof generation dominates latency at 64–85% of total cycle time across all tested backends. Pipelined execution achieves a 1.29× throughput improvement over sequential processing for 64-transaction batches with the rapidsnark backend, yielding 4.37 tx/s sustained throughput. The results identify a critical constraint: the snarkjs WASM prover exceeds the 90-second target at batch size 64 (156.9 s per batch), requiring either a native prover (rapidsnark: 18.9 s) or batch size reduction to 16 (31.4 s) for production deployment. All findings are parameterized for downstream TLA+ formalization and implementation.**

***Index Terms*—ZK validium, node orchestration, pipelined state machine, enterprise blockchain, zero-knowledge proofs, Groth16, Avalanche**

I. Introduction

Enterprise blockchain deployments in regulated industries—industrial traceability, supply chain management, financial record-keeping—face a fundamental architectural challenge: how to orchestrate the complete lifecycle of a zero-knowledge validity proof while maintaining continuous transaction availability and strict data privacy. Unlike public rollup sequencers where proof generation latency is amortized across thousands of users, enterprise validium nodes serve a single organization with deterministic performance requirements and zero tolerance for data leakage [1].

Production ZK L2 systems such as Polygon zkEVM [2], [3], zkSync Era [4], [5], and Scroll [6], [7] decompose node operations into specialized components—sequencers, aggregators, provers, relayers—connected by internal queues. Recent work on orchestration frameworks, particularly the push0 system deployed on Zircuit [8], demonstrates that event-driven orchestration with message bus coordination achieves sub-10 ms overhead with fault-tolerant recovery. However, these architectures target public L2 infrastructure with horizontally scaled prover pools and are over-provisioned for enterprise validium deployments where a single node serves one organization.

This paper addresses the gap between public L2 orchestration patterns and enterprise validium requirements. We present the design of a pipelined, event-driven state machine orchestrator for the Basis Network enterprise ZK validium node deployed on Avalanche. The orchestrator integrates five previously characterized components—Sparse Merkle Tree (RU-V1), ZK state transition circuit (RU-V2), L1 state commitment contract (RU-V3), crash-safe batch aggregation (RU-V4), and Shamir-based data availability committee (RU-V6)—into a single Node.js service that processes the complete receive-to-settlement cycle.

Our contributions are threefold. First, we define a typed finite state machine with six states and 17 transitions that supports pipelined execution: the node continues ingesting transactions while a proof is being generated or submitted. Second, we provide comprehensive benchmarks demonstrating that orchestration overhead is 593 ms per batch (0.66% of the 90-second budget), with proof generation consuming 64–85% of end-to-end latency across all tested configurations. Third, we quantify the pipeline speedup (1.03–1.29×) as a function of batch size and prover backend, establishing that the speedup increases with batch size because witness generation time grows relative to proving time, enabling greater overlap between preparation and proving phases. These results, together with six formal invariants for downstream TLA+ specification, provide a complete blueprint for enterprise ZK validium node implementation.

II. Related Work

A. Production ZK Node Architectures

The architecture of ZK L2 nodes has converged toward a multi-component pattern where transaction sequencing, batch formation, proof generation, and L1 submission are handled by separate processes.

Polygon zkEVM [2] decomposes its node into a Sequencer (transaction ordering), Aggregator (proof orchestration), zkProver (recursive STARK-to-SNARK proving), and

Synchronizer (L1 state recovery). The Aggregator delegates proof tasks to the zkProver and manages retries on failure. Hermez 2.0 [3] introduced the Proof of Efficiency consensus mechanism, establishing the batch lifecycle pattern (sequence, aggregate, verify) that subsequent systems adopted.

zkSync Era [4], [5] separates execution from proving: the Sequencer processes transactions and produces soft confirmations (sub-second), while the Batcher groups blocks into batches for proving on a separate timeline. Recovery is idempotent—the Batcher resumes from the first uncommitted batch. This architecture explicitly decouples user-facing latency from proof generation latency.

Scroll [6], [7] introduces a two-stage proving pipeline: blocks are grouped into chunks, chunks are proved individually, and chunk proofs are aggregated into batch proofs. A Coordinator randomly assigns chunk proofs to a pool of provers, providing horizontal scalability. Failed proof tasks are reassigned to other provers.

B. Event-Driven ZK Orchestration

The push0 framework [8], deployed on Zircuit since March 2025, represents the state of the art in ZK proof orchestration. push0 uses stateless dispatchers and collectors coordinated via NATS JetStream, achieving P50 orchestration latency of 5.2–5.3 ms and P99 of 8.0–8.5 ms on a 3-node Kubernetes cluster. Orchestration overhead is 0.05–0.1% of proving time. Crash recovery achieves 0 task loss for dispatchers and 0.5 s mean time to recovery (MTTR) for collectors. The system scales to 32 concurrent dispatchers at 99% efficiency.

Table I compares the architectural patterns across production systems and our proposed design.

C. State Machine Design Patterns

Petrasch [9] describes the transformation of state machines to event-driven architectures in enterprise systems, establishing the pattern of typed state enumerations with explicit transition guards. The Saga Orchestrator pattern [10] applies state machines to long-running transactions with compensating actions, directly analogous to the ZK proof pipeline where a failed proving step must roll back batch state. Event sourcing with the Decider pattern [11] provides a command-to-event model for TypeScript implementations.

D. Crash Recovery and Persistence

Write-ahead logging (WAL) is the foundational crash recovery mechanism for database systems. Mohan et al. [12] established the ARIES protocol with checkpoint-based recovery and log sequence number (LSN) tracking. Modern implementations for Node.js applications [13] and graceful shutdown patterns [14] ensure that in-flight operations are persisted before process termination. Our prior work (RU-V4) demonstrated WAL-backed crash recovery for ZK batch aggregation with sub-150 μs write latency and zero transaction loss [15].

E. ZK Prover Performance

Groth16 [16] remains the standard for succinct ZK proofs with constant-size verification. Prover performance varies by implementation: snarkjs (WASM) [17] achieves 12,757 ms for 274K constraints on commodity hardware, while rapidsnark (C++ with native field arithmetic) offers 4–10$\times$ speedup [18]. ICICLE-Snark (GPU-accelerated) achieves 63$\times$ MSM improvement [19], reducing proving time to 1–3 s for circuits in the 1–3M constraint range. The ZK proving infrastructure landscape [20] shows a trend toward prover decentralization and hardware acceleration.

F. HTTP Framework Selection

For the enterprise node API layer, framework selection directly impacts throughput. Empirical comparisons [21] demonstrate that Fastify achieves 2–4$\times$ higher request throughput than Express.js with native JSON schema validation and lower memory overhead. Fastify's WebSocket integration [22] enables real-time event streaming for enterprise client applications.

III. System Model

A. Network Architecture

The Basis Network enterprise ZK validium operates on a three-tier architecture:

1) **Enterprise Systems Layer**: Client applications (PLASMA for industrial maintenance, Trace for SME ERP) that produce business transactions. These systems hash all data client-side before submission; the node never receives raw enterprise data.
2) **Validium Node Layer**: A single Node.js service per enterprise that maintains off-chain state (Sparse Merkle Tree), aggregates transactions into batches, generates Groth16 ZK proofs, distributes witness data to a Data Availability Committee (DAC), and submits proofs to the settlement layer.
3) **Settlement Layer**: The Basis Network L1, an Avalanche Subnet-EVM chain (Chain ID 43199) with zero-fee permissioned transactions and sub-second Snowman consensus finality.

B. Trust Assumptions

- The enterprise node operator is trusted for liveness but not for data integrity (proofs enforce correctness).
- The DAC operates under a t-of-n threshold (default 2-of-3): data remains available if at least t committee members are honest.
- The L1 settlement layer is assumed correct (Avalanche consensus with permissioned validator set).
- Enterprise client systems are trusted to correctly hash data before submission (the node verifies signatures but not data content).

TABLE I
PRODUCTION ZK NODE ARCHITECTURE COMPARISON

System	Architecture	Proving Model	Batch Pipeline	Crash Recovery	Scale Target
Polygon zkEVM	Seq + Agg + zkProver	Recursive STARK→SNARK	Seq forms, Agg proves	Sync replays from L1	Public L2
zkSync Era	Seq + Batcher + Prover	Execution ≠ proving	Blocks→Batches→L1	Idempotent resume	Public L2
Scroll	Seq + Coord + Pool	Coord assigns to pool	Blocks→Chunks→Batches	Reassign failed tasks	Public L2
push0/Zircuit	Dispatchers + Bus	Stateless dispatchers	Ranges→Recursive→Groth16	At-least-once replay	Public L2
Ours	**Single node, 3 loops**	**Child process**	**Txs→Batch→Prove→Submit**	**WAL + checkpoint**	**Enterprise**

C. Privacy Model

The zero data leakage invariant requires that no raw enterprise data crosses the node boundary. The data flow is:

1) Enterprise systems compute $h = \text{SHA-256}(\text{record data})$ and submit transaction tuples (k, h, σ) where k is a deterministic key derived from the record identifier and σ is the enterprise signature.
2) The node inserts (k, h) into the Sparse Merkle Tree, generating state root transitions $r_{\text{prev}} \to r_{\text{new}}$.
3) The ZK proof reveals only: r_{prev}, r_{new}, batch number, and enterprise identifier. No transaction keys, values, or counts are disclosed.
4) Witness data is distributed to the DAC via Shamir secret sharing (information-theoretically secure: no single committee member can reconstruct the witness).
5) The L1 contract stores only: state roots, proofs, and batch metadata.

D. Performance Requirements

The orchestrator must satisfy:

- **End-to-end latency**: < 90 s for a batch of 64 transactions (60 s proving budget + 30 s overhead budget).
- **Continuous availability**: The node must accept new transactions during proof generation and L1 submission (no downtime windows).
- **Crash recovery**: Zero transaction loss under arbitrary crash scenarios, with bounded recovery time.
- **Memory footprint**: < 500 MB for a single enterprise node with up to 100K state entries.

IV. METHODOLOGY

A. State Machine Design

The orchestrator is modeled as a typed finite state machine $\mathcal{M} = (S, E, \delta, s_0)$ where S is the set of states, E is the set of events, $\delta : S \times E \to S$ is the transition function, and $s_0 = \text{Idle}$ is the initial state.

1) States: The state set $S = \{\text{Idle}, \text{Receiving}, \text{Batching}, \text{Proving}, \text{Submitting}, \text{Error}\}$ captures the six phases of the validium proof cycle:

- **Idle**: No pending work. The node polls for new transactions.
- **Receiving**: Actively accepting transactions from enterprise systems via REST or WebSocket API.
- **Batching**: Batch threshold reached. The node forms a batch, applies state updates to the SMT, and generates the circuit witness.
- **Proving**: ZK proof generation in progress. Runs in a child process to avoid blocking the event loop.
- **Submitting**: Proof generated. DAC attestation and L1 settlement in progress.
- **Error**: Recoverable error state. Transitions back to Idle via retry with exponential backoff.

2) Transition Table: Table II enumerates all 17 valid transitions. The critical design feature is pipelined receiving: the TRANSACTIONRECEIVED event is accepted in the Proving and Submitting states, enabling continuous ingestion while a proof cycle completes.

TABLE II
STATE MACHINE TRANSITION TABLE

Source	Event	Target
Idle	TransactionReceived	Receiving
Idle	ShutdownRequested	Idle
Receiving	TransactionReceived	Receiving
Receiving	BatchThresholdReached	Batching
Receiving	ErrorOccurred	Error
Batching	WitnessGenerated	Proving
Batching	ErrorOccurred	Error
Proving	ProofGenerated	Submitting
Proving	TransactionReceived	Proving
Proving	ErrorOccurred	Error
Submitting	BatchSubmitted	Submitting
Submitting	L1Confirmed	Idle
Submitting	TransactionReceived	Submitting
Submitting	ErrorOccurred	Error
Error	RetryRequested	Idle
Error	ShutdownRequested	Idle

3) Invariants: Six invariants govern correctness and are formalized for downstream TLA+ specification:

INV-NO1 *Liveness*: If $|\text{pendingTx}| > 0$ and $s = \text{Idle}$, then eventually $s = \text{Proving}$.

INV-NO2 *Safety*: If $s = \text{Submitting}$, then $\text{proof.prevRoot} = \text{smt.prevRoot}$ and $\text{proof.newRoot} = \text{smt.currentRoot}$.

INV-NO3 *Privacy*: The only data transmitted outside the node boundary are: proof (a, b, c), public signals $(r_{\text{prev}}, r_{\text{new}}, \text{batchNum}, \text{enterpriseId})$, and Shamir shares to DAC nodes.

INV-NO4 *Crash Recovery*: After crash and restart, $|\text{walReplayed}| + |\text{committed}| = |\text{totalEnqueued}|$.

INV-NO5 *State Root Continuity*: For batch N, $\text{submit}(N).\text{prevRoot} = \text{batch}(N-1).\text{newRoot}$.

INV-NO6 *Single Writer*: Only the batch loop modifies the SMT. No concurrent write access is permitted.

B. Pipelined Architecture

The sequential execution model (Idle → Receive → Batch → Prove → Submit → Idle) blocks transaction ingestion during proof generation, creating unavailability windows of 12+ seconds. Following the zkSync Era [4] and push0 [8] patterns, we decompose the pipeline into three concurrent loops:

1) **Ingestion Loop**: Continuously accepts transactions via the Fastify REST/WebSocket API, writes to the WAL, and enqueues to the transaction queue.
2) **Batch Loop**: Monitors the queue size. When the batch threshold is reached, dequeues transactions, applies SMT updates, and generates the circuit witness.
3) **Proving/Submission Loop**: Takes formed batches, invokes the ZK prover (child process), distributes witness data to the DAC, and submits the proof to L1.

The pipelined speedup arises when the batch loop prepares the next batch while the proving loop processes the current one. The speedup is bounded by:

$$\text{Speedup} = \frac{N \cdot T_{\text{seq}}}{T_{\text{cold}} + (N-1) \cdot \max(T_{\text{prep}}, T_{\text{prove+submit}})} \quad (1)$$

where T_{seq} is the sequential batch time, T_{cold} is the first batch (no overlap), T_{prep} is preparation time (batch formation + witness generation), and $T_{\text{prove+submit}}$ is the proving and submission phase. The speedup increases with batch size because T_{prep} grows (more witness generation work) while $T_{\text{prove+submit}}$ grows more slowly.

C. Component Integration

The orchestrator integrates five components, each characterized by prior research units in this pipeline:

TABLE III
COMPONENT LATENCIES FROM PRIOR RESEARCH

Component	Operation	Latency	Source
SparseMerkleTree	Insert (d32)	1.825 ms	RU-V1
SparseMerkleTree	Proof generation	0.018 ms	RU-V1
TransactionQueue	WAL write (group)	149 μs	RU-V4
BatchAggregator	Batch formation	0.02 ms	RU-V4
BatchBuilder	Witness gen (d32, b8)	578 ms	RU-V2
DACProtocol	Full attestation	163 ms	RU-V6
StateCommitment	L1 submission	268K gas	RU-V3

D. API Design

The node exposes a Fastify-based REST API and WebSocket event stream for enterprise client integration:

- `POST /v1/transactions` – Submit a single transaction
- `POST /v1/transactions/batch` – Submit multiple transactions
- `GET /v1/status` – Node health and current state
- `GET /v1/batches/:id` – Query batch status and proof
- `GET /v1/state/:enterprise` – Current state root

WebSocket events (`ws://node/v1/events`) provide real-time notifications for transaction acceptance, batch proving progress, and L1 confirmation.

E. Technology Selection

Fastify v4 was selected over Express.js based on published benchmarks [21] showing 2–4× higher request throughput with native TypeScript support and JSON schema validation. WebSocket integration uses the `@fastify/websocket` [22] plugin for real-time enterprise event streaming.

V. EXPERIMENTAL SETUP

A. Benchmark Methodology

The experimental evaluation consists of two complementary benchmarks: (1) an *orchestration overhead benchmark* that measures per-phase latencies across the complete proof cycle, and (2) a *pipeline comparison benchmark* that quantifies the throughput improvement of pipelined over sequential execution.

1) Orchestration Overhead Benchmark: Five scenarios are evaluated, each varying the batch size and prover backend while holding the circuit depth constant at 32 levels:

1) `orchestration_overhead_only`: Batch 8, 0 ms proving (isolates pure orchestration cost). 50 iterations.
2) `snarkjs_d32_b8`: Batch 8, 12,757 ms proving (snarkjs WASM, 274K constraints). 30 iterations.
3) `rapidsnark_d32_b8`: Batch 8, 2,500 ms proving (rapidsnark, 274K constraints). 30 iterations.
4) `snarkjs_d32_b16`: Batch 16, 28,000 ms proving (snarkjs WASM, ∼548K constraints). 30 iterations.
5) `rapidsnark_d32_b64`: Batch 64, 12,000 ms proving (rapidsnark, ∼2.2M constraints). 30 iterations.

Each scenario executes 3 warm-up iterations (discarded) followed by the measurement iterations. Per-phase timings are recorded for: batch formation, witness generation, proving, DAC attestation, L1 submission, and checkpoint. Orchestration overhead is computed as the sum of batch formation and witness generation.

2) Pipeline Comparison Benchmark: Six configurations are simulated with 10 sequential batches each, comparing sequential and pipelined architectures. Component latencies are calibrated from prior research unit measurements:

- SMT insert: 1.8 ms/tx (RU-V1, Poseidon hash, depth 32)
- WAL write: 0.149 ms/tx (RU-V4, group commit)
- Batch formation: 0.02 ms (RU-V4)
- Witness generation: 72 ms/tx (RU-V2, 578 ms for 8 txs)
- DAC attestation: 163 ms (RU-V6, JavaScript, 500 KB)
- L1 submission: 2,000 ms (estimated Avalanche Fuji block time)

Proving times are parameterized per backend: snarkjs (12,757 ms for b8, 28,000 ms for b16, 150,000 ms for b64)

and rapidsnark (2,500 ms for b8, 5,000 ms for b16, 12,000 ms for b64).

B. Mock Calibration

Stage 1 benchmarks use mock components calibrated to measured latencies from prior experiments to isolate orchestration overhead from component variability. Each mock implements the same TypeScript interface as the production component:

- **MockSMT**: Simulates Poseidon hash latency via CPU-bound SHA-256 iterations (32 rounds per insert) with spin-wait padding to match the 1.0–1.8 ms target from RU-V1.
- **MockBatchBuilder**: Simulates witness generation at 72 ms/tx, scaling linearly with batch size. Calibrated to the 578 ms measurement from RU-V2 for depth-32, batch-8 configurations.
- **MockProver**: Configurable timer-based delay returning a valid Groth16 proof structure. Default: 12,757 ms (snarkjs d32, b8).
- **MockDAC**: 163 ms delay matching the JavaScript Shamir attestation measurement from RU-V6.
- **MockL1Submitter**: 2,000 ms delay matching the estimated Avalanche Fuji block confirmation time.

C. Statistical Protocol

For each scenario, we report mean, standard deviation, minimum, maximum, and P95 latency. The 95% confidence interval (CI) half-width must be within 10% of the mean for the result to be accepted. All replications use independent random seeds for transaction generation.

D. Platform

All benchmarks execute on Node.js v22 (Windows 11, x64) with commodity hardware. Memory footprint is measured via `process.memoryUsage().heapUsed`.

E. State Machine Validation

A dedicated test suite validates the state machine implementation with 46 test cases covering: happy path transitions, pipelined receiving during Proving and Submitting, error recovery from all states, invalid transition rejection, transition history tracking, and event emitter correctness.

VI. Results

A. State Machine Validation

All 46 test cases pass, validating the state machine implementation across six categories: happy path cycle (6 transitions from Idle through Submitting back to Idle), pipelined receiving (transaction acceptance during Proving and Submitting states), error recovery (all operational states transition to Error; Error recovers to Idle), invalid transition rejection (descriptive errors for undefined transitions), history tracking (full audit trail with metadata), and event emitter correctness (per-transition and per-state events).

B. Orchestration Overhead

Table IV presents per-phase latencies across all five benchmark scenarios.

1) Phase Breakdown: Table V details the per-phase timings for the two primary scenarios: batch-8 overhead isolation and the target batch-64 configuration.

2) Overhead Scaling: Orchestration overhead scales linearly with batch size at a rate of approximately 73.5 ms per transaction, dominated by witness generation (72 ms/tx). Batch formation contributes 1.4 ms/tx. The linear scaling relationship is:

$$T_{\text{overhead}}(n) = 11.4 + 72.7 \cdot n \quad (\text{ms}, n = \text{batch size}) \tag{2}$$

verified at $n \in \{8, 16, 64\}$: predicted values of 593, 1,175, and 4,664 ms match measured values of 593, 1,187, and 4,704 ms within 1%.

3) Statistical Quality: For the batch-8 overhead scenario (50 iterations): mean 2,784.8 ms, standard deviation 7.8 ms, 95% CI half-width 2.2 ms (0.08% of mean). All scenarios achieve CI half-width <1% of mean, well below the 10% acceptance threshold.

C. Pipeline Speedup

Table VI compares sequential and pipelined execution across six configurations with 10 batches each.

1) Speedup Analysis: The pipeline speedup is a function of the ratio $\rho = T_{\text{prep}}/T_{\text{prove+submit}}$. When preparation time is negligible relative to proving ($\rho \to 0$), the pipeline offers no benefit (speedup $\to$ 1.0). As ρ increases, more preparation work overlaps with the proving phase:

- snarkjs b8: $\rho = 591/14{,}920 = 0.04$, speedup 1.04×
- rapidsnark b8: $\rho = 591/4{,}670 = 0.13$, speedup 1.11×
- rapidsnark b64: $\rho = 4{,}733/14{,}170 = 0.33$, speedup 1.29×

The maximum theoretical speedup (unlimited batch size, equal preparation and proving time) approaches 2.0× but is constrained by the proving phase floor.

D. End-to-End Latency vs. Target

Table VII evaluates each configuration against the 90-second target.

The snarkjs WASM backend at batch size 64 exceeds the target by 1.7× (156.9 s vs. 90 s). All other configurations meet the target with margins ranging from 2.9× to 17.0×.

E. Memory Footprint

Heap memory usage is stable across all scenarios at 84.8–87.1 MB. The 2.3 MB variance corresponds to the additional transaction objects held in memory for larger batch sizes. Production deployment with a real SMT containing 100K entries is estimated at ∼320 MB (234 MB SMT from RU-V1 + 85 MB orchestrator), within the 500 MB target.

TABLE IV
ORCHESTRATION BENCHMARK RESULTS (30–50 ITERATIONS PER SCENARIO, DEPTH 32)

Scenario	Batch Size	Total E2E (ms)	Overhead (ms)	Proving (ms)	Overhead/E2E (%)	Memory (MB)
overhead_only	8	2,785	593	12†	21.3‡	84.8
snarkjs d32 b8	8	15,537	594	12,765	3.8	85.6
rapidsnark d32 b8	8	5,281	594	2,507	11.2	86.2
snarkjs d32 b16	16	31,371	1,187	28,008	3.8	85.9
rapidsnark d32 b64	64	18,889	4,704	12,006	24.9	87.1

†Timer overhead only (0 ms simulated proving). ‡Excluding proving time; overhead as fraction of non-proving E2E.

TABLE V
PER-PHASE LATENCY BREAKDOWN

Phase	Batch 8		Batch 64	
	Mean (ms)	P95 (ms)	Mean (ms)	P95 (ms)
Batch formation	11.39	12.67	90.31	92.43
Witness generation	581.86	584.20	4,613.21	4,622.13
Proving	12.31	15.98	12,006.26	12,016.35
DAC attestation	171.83	174.23	171.89	174.97
L1 submission	2,007.26	2,015.60	2,007.35	2,015.34
Checkpoint	<0.01	<0.01	<0.01	<0.01
Total	**2,784.8**	–	**18,889.2**	–

TABLE VI
SEQUENTIAL VS. PIPELINED ARCHITECTURE (10 BATCHES)

Configuration	Seq. (s)	Pipe. (s)	Speed-up	Tput (tx/s)
snarkjs b8	155.1	149.8	1.04×	0.53
rapidsnark b8	52.5	47.2	1.11×	1.69
snarkjs b16	313.5	302.8	1.04×	0.53
rapidsnark b16	83.5	72.8	1.15×	2.20
snarkjs b64	1,569	1,526	1.03×	0.42
rapidsnark b64	**189.0**	**146.4**	**1.29×**	**4.37**

TABLE VII
END-TO-END LATENCY VS. 90-SECOND TARGET

Backend	Batch	E2E (s)	Under 90s?	Margin
snarkjs	8	15.5	Yes	5.8×
rapidsnark	8	5.3	Yes	17.0×
snarkjs	16	31.4	Yes	2.9×
rapidsnark	16	9.2	Yes	9.8×
snarkjs	**64**	**156.9**	**No**	**1.7× over**
rapidsnark	64	18.9	Yes	4.8×

TABLE VIII
BENCHMARK RECONCILIATION WITH PUBLISHED DATA

Metric	Ours	Published	Assessment
Orch. overhead	593 ms	push0: 5.2 ms	Expected*
Batch formation only	11.4 ms	push0: 5.2 ms	2.2× (consistent)
Memory per component	85 MB	push0: 8–14 MB	Expected†

*push0 measures message routing only, not witness generation.
†push0 dispatchers are stateless; our node holds full SMT state.

F. Benchmark Reconciliation

Table VIII validates our measurements against published data.

Batch formation latency (11.4 ms) is 2.2× the push0 P50 (5.2 ms), consistent with our node performing 8 SHA-256 hash computations for mock SMT inserts while push0 dispatchers perform message routing only. No metric diverges by more than 10× from published benchmarks after accounting for architectural differences.

VII. DISCUSSION

A. Hypothesis Evaluation

The hypothesis stated that a Node.js event-driven orchestrator can process a 64-transaction batch in under 90 seconds with zero data leakage and crash recovery. Stage 1 results provide partial confirmation:

- **Orchestration overhead**: 4,704 ms for batch 64, well below the 30-second budget (H0 rejected with 6.4× margin).
- **E2E latency with rapidsnark**: 18.9 s (sequential) or 14.6 s (pipelined), meeting the 90 s target with 4.8–6.2× margin.
- **E2E latency with snarkjs**: 156.9 s at batch 64, exceeding the target by 1.7×. H0 cannot be rejected for the WASM backend at this batch size.
- **State machine correctness**: 46/46 tests pass, pipelined model validated.
- **Crash recovery**: Design supports zero-loss recovery via WAL + checkpoint; empirical validation deferred to Stage 3.
- **Privacy**: Zero data leakage by construction (node receives only hashes); formal audit deferred to Stage 3.

B. The Proving Bottleneck

Across all configurations, ZK proof generation dominates end-to-end latency, consuming 64% (rapidsnark b64) to 85% (snarkjs b8) of the total cycle time. Orchestration overhead—the sum of batch formation, witness generation, state management, and scheduling—is a negligible fraction of the proving budget. This finding is consistent with the push0 architec-

ture [8], which reports orchestration overhead of 0.05–0.1% of proving time.

The practical implication is that node performance optimization should focus exclusively on prover selection and circuit optimization, not on orchestration layer engineering. A 2× improvement in orchestration overhead (from 593 ms to 296 ms) saves 297 ms, while a 2× improvement in proving (from 12,757 ms to 6,378 ms) saves 6,379 ms—a 21× greater impact.

C. Prover Backend Constraint

The snarkjs WASM backend, while sufficient for batch sizes up to 16, cannot achieve the 90-second target at batch 64. Three mitigation strategies are available, ordered by recommendation:

1) **Rapidsnark** (recommended for production): C++ native prover achieving 4–10× speedup over snarkjs, reducing batch-64 proving from ∼150 s to ∼12 s. Requires native binary installation.
2) **Batch size reduction** (safest for MVP): Reducing batch size to 16 keeps snarkjs within 31.4 s E2E, at the cost of lower throughput (0.53 tx/s vs. 4.37 tx/s).
3) **GPU prover** (future): ICICLE-Snark [19] achieves 63× MSM improvement, potentially reducing batch-64 proving to 1–3 s. Requires GPU infrastructure.

D. Pipeline Speedup Characteristics

The 1.29× speedup at batch 64 with rapidsnark is modest but meaningful: it reduces 10-batch processing from 189 s to 146 s, saving 43 s. The speedup is inherently bounded because the proving phase dominates—even with perfect overlap, preparation cannot be fully hidden behind proving when $T_{\text{prep}} < T_{\text{prove+submit}}$.

The speedup would increase under two conditions: (1) faster provers that reduce $T_{\text{prove+submit}}$ closer to T_{prep}, or (2) larger batch sizes that increase T_{prep} via witness generation. With GPU provers (1–3 s proving), preparation time (4.7 s for b64) would exceed proving time, making the preparation phase the bottleneck and increasing speedup toward 2.0×.

E. Comparison with Production Systems

Our single-node architecture differs fundamentally from production L2 systems (Table I) in scale target: Polygon, zkSync, and Scroll serve millions of users with horizontally scaled prover pools, while our node serves a single enterprise. This simplification eliminates the need for prover coordination, task scheduling, and consensus mechanisms, but preserves the core orchestration challenge: managing the asynchronous proof lifecycle while maintaining continuous availability.

The push0 comparison is most informative. Our batch formation latency (11.4 ms) is 2.2× push0's P50 (5.2 ms), attributable to our node performing actual state tree operations (8 hash computations) versus push0's pure message routing. When comparing equivalent operations (message routing vs. batch formation without SMT), the overhead is comparable.

F. Limitations

1) **Mock components**: Stage 1 uses calibrated mocks, not real components. Real SMT, snarkjs, and ethers.js may introduce variability not captured by deterministic mocks. Stage 2 will replace all mocks.
2) **Single enterprise**: All benchmarks use a single enterprise identifier. Concurrent enterprises sharing a node may introduce contention on the SMT and queue.
3) **Crash recovery not tested**: The WAL + checkpoint design is described but not empirically validated. Stage 3 will inject crashes at each pipeline phase.
4) **No network latency**: L1 submission uses a 2 s mock. Actual Avalanche Fuji latency may vary with network conditions.
5) **Pipeline simulation**: The pipeline comparison uses analytical simulation, not concurrent thread execution. Real pipelining may encounter event loop contention in Node.js.

VIII. Conclusion

We presented the design and empirical evaluation of a pipelined, event-driven state machine orchestrator for enterprise ZK validium nodes. The orchestrator implements six states with 17 transitions, enabling concurrent transaction ingestion during proof generation through three decoupled processing loops.

The key findings are:

1) **Orchestration overhead is negligible**: At 593 ms per 8-transaction batch (0.66% of the 90-second budget), the orchestration layer—including batch formation, SMT state updates, and witness generation—adds minimal overhead to the proof cycle. The overhead scales linearly at 73.5 ms/tx, entirely dominated by witness generation.
2) **Proving dominates latency**: ZK proof generation consumes 64–85% of end-to-end cycle time across all tested configurations. Performance optimization efforts should target prover backend selection and circuit constraint reduction, not orchestration engineering.
3) **Pipelining improves throughput modestly**: The 1.29× speedup at batch 64 with rapidsnark reduces 10-batch processing by 43 s. The speedup scales with the preparation-to-proving time ratio and would increase with faster provers or larger batches.
4) **Prover backend determines feasibility**: The snarkjs WASM backend fails the 90-second target at batch 64 (156.9 s), while rapidsnark meets it with 4.8× margin (18.9 s). Production deployment requires a native or GPU-accelerated prover for batch sizes above 16.

The six formal invariants (liveness, safety, privacy, crash recovery, state root continuity, single writer) provide a complete specification for downstream TLA+ formalization. The component integration map—spanning five prior research units covering SMT, circuits, L1 contracts, batch aggregation, and data availability—establishes the implementation blueprint for the Basis Network enterprise validium node.

Future work will replace mock components with production implementations (Stage 2), validate crash recovery under fault injection (Stage 3), and ablate individual components to quantify their marginal contribution to system latency (Stage 4).

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